Near Real-Time Burned Area Progression Mapping With Multispectral Data Using Ensemble Learning

Xikun Hu, Hao Wen, Puzhao Zhang, Ka-Veng Yuen, and Ping Zhong, *Senior Member, IEEE*

***Abstract*— Monitoring the wildfire progression is essential to quantify the fire-disturbance areas for emergency responses. To combine the advantages of pixelwise machine learning (ML) method and region-based deep learning (DL) segmentation model, this study proposes a two-phase hybrid framework for near real-time burned area progression mapping: the first one intends to depict burned area delimitation using a contextual algorithm HRNet to exclude the unburned areas outside the perimeter and minimize omission errors, which partially remain unburned patches within the delimitation as commission errors. The second phase refines the burned area spatially using ensemble fusion based on an updating support vector machine (SVM) model under the voting scheme as new imagery arrives to reduce the commission errors consecutively. The validation results showed that the accuracy of perimeter prediction using the HRNet can reach 96.77% in Kappa. The iterative optimization can improve the average Kappa value from 62.55% to 70.75% for burned area pixel classification using pixelwise SVM alone. The proposed ensemble learning framework can further refine the burned area progression results, reaching an average Kappa up to 85.19%, at four acquisition dates with Sentinel-2 and Landsat-8 available during the Sand fire event that occurred in California.**

***Index Terms*— Burned area mapping, deep learning (DL), multispectral data, near real time, wildfire.**

I. Introduction

NATURAL wildfires make significant impacts on humans and the environment directly or indirectly, including residential areas, human lives, infrastructure, vegetation, ecosystem cycles, and biodiversity [1], [2]. Latest research has shown that drought causes overnight fires in North America, as a major driver of large wildfires [3]. Accurate burned area progression mapping during the wildfire plays an important role in damage prevention and prediction, to help monitor the dynamic changes and minimize losses.

Earth observation has provided a remote sensing approach for monitoring wildfire under different spatial and temporal resolutions [4]. Coarse-resolution satellites, such as Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS), successfully detect the active fire hotspots with daily temporal resolution but low spatial resolution at either 1000 or 375 m [5]. Medium-resolution sensors (e.g., Landsat-8 and Sentinel-2), on the other hand, support more precise detection of burned areas at regional and local scales with a global median average revisit 2.9 days in near real time [6]. The collaborative use of multiple satellite sources and taking advantage of their complementarity show potential in mapping the burned area progression [7]. However, how to make full use of multispectral satellite data in time series remains challenging, while maintaining reliability and robustness.

Regarding the burned area progression mapping, the provisional burned area products were derived using the delta normalized burn ratio (dNBR) threshold to generate intra-year metrics of large wildfires throughout a fire season [8]. Many machine learning (ML) algorithms have been used for burned area mapping [9], including logical regression, support vector machines (SVMs), AdaBoost, and self-organizing map [10], [11], [12]. The random forest algorithm is applied to combine the polarization and texture features of satellite SAR images to extract the burned areas [13]. SVM has been widely used for both classification and regression problems in fire-related research [14], [15], and it shows good performances with limited training samples. Zammit et al. [16] compared SVM, K-nearest neighbors, and the K-means algorithm for burned area mapping over two wildfire events in France, and SVM showed the best overall performance. Recently, deep learning (DL)-based models have been widely used in burned area detection using abundant medium-resolution satellite data [17], [18], [19]. Burnt-Net was proposed for burned area detection based on a morphological learnable deep-learning model with postfire Sentinel-2 imagery [20]. A novel learning-without-forgetting framework was proposed for continuous wildfire progression mapping using multisensor data [21]. A CNN-based online learning framework was applied to Sentinel-1 SAR time series for wildfire monitoring [22].

The algorithms mentioned above focus on either ML-based or DL-based models. ML-based methods show high efficiency but low estimation accuracy with huge commission errors outside the fire perimeters. Conversely, DL-based models have good performance in object-based representations for burned area delineations but show a few commission errors inside the perimeters. To find a trade-off, in this study, we combine

Manuscript received 22 March 2024; revised 30 May 2024; accepted 6 June 2024. Date of publication 11 June 2024; date of current version 19 June 2024. This work was supported by the National Natural Science Foundation of China under Grant 62301574. *(Corresponding author: Ping Zhong.)*

Xikun Hu, Hao Wen, and Ping Zhong are with the National Key Laboratory of Automatic Target Recognition, College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: xikun@nudt.edu.cn; hao.wen@nudt.edu.cn; zhongping@nudt.edu.cn).

Puzhao Zhang is with the Key Laboratory of Collaborative Intelligence Systems of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: puzhao@kth.se).

Ka-Veng Yuen is with the State Key Laboratory on Internet of Things for Smart City, Department of Civil and Environmental Engineering, University of Macau, Macau, China (e-mail: kvyuen@um.edu.mo).

Digital Object Identifier 10.1109/LGRS.2024.3412173

1558-0571 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:30:39 UTC from IEEE Xplore. Restrictions apply.

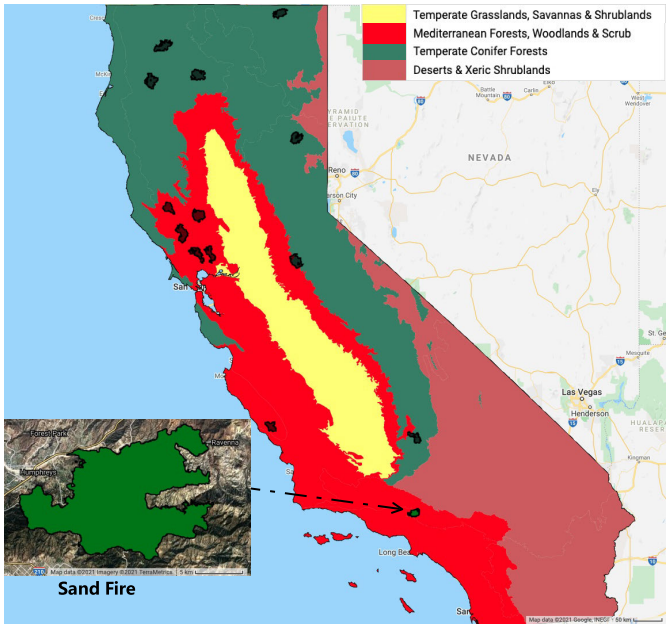

Fig. 1. Thirteen wildfire events for training are masked with black polygons, and the illustrative Sand fire is pointed by a dashed-dotted line. The base map is the RESOLVE Ecoregions from Google Earth Engine, adapted from [24].

the advantages of the ML-based SVM model for classifying burned area pixels and the DL-based segmentation model HRNet for extracting the boundary delineations. This ensemble learning framework refines the burned areas continuously by employing the pixelwise SVM model whose parameters are updated based on previous results, and the region-based HRNet initially trained well [23]. We collect the multispectral imageries over 13 large wildfire events across CA, USA, for initial training of SVM and HRNet, and validate the performance of progression mapping using Sentinel-2 and Landsat-8 data available during the Sand fire event.

II. Study Areas and Data Characteristics

A. Study Areas

California experienced large numbers of wildfires every year, which resulted in dramatic property losses and infrastructure damage. In this study, the reference data originate from the historic fire perimeters of the California Department of Forestry and Fire Protection (CalFire). Fig. 1 shows 14 extreme wildfire events with CalFire perimeter data that burned over 40 000 acres between 2015 and 2019. Among these wildfire events, the Sand fire, near the city of Santa Clarita, was taken as a test scenario to illustrate the performance of our proposed method in monitoring burned area progression. The Sand fire is a large long-duration wildfire over an actively burning period of more than two weeks, during which four cloudless multispectral imageries were collected from Sentinel-2 and Landsat-8. The postfire multispectral imageries over the remaining 13 wildfire events are used for training the initial SVM and HRNet models, together with the reference data from CalFire.

B. Multispectral Data

Twin satellites Sentinel-2A and Sentinel-2B carry the MultiSpectral Instrument (MSI) that samples 13 spectral bands with spatial resolution ranging from 10 to 60 m. Operational land imager (OLI) sensor in Landsat-8 offers 16-day repetitive Earth coverage and collects image data for nine spectral bands with a 30-m spatial resolution except for the 15-m panchromatic band. Sentinel-2 MSI has higher temporal resolution (5 versus 16 days) and spatial resolution (10/20 versus 30 m). It is worth noting that Sentinel-2 MSI B8A rather than B8 is quite similar to the OLI B5, and their SWIR bands have a similar bandwidth specification [25]. In this study, Sentinel-2 MSI top of atmosphere (TOA) and Landsat-8 OLI TOA reflectance data were extracted before, during, and after the Sand fire event. All multispectral images were downloaded based on Google Earth Engine and resampled into 20 m. In addition, the preprocessing operations, such as date filtering, bounds filtering to the ROIs, and subsets clipping, were conducted. Three fire-sensitive channels B12, B11, and B8A of Sentinel-2 and B7, B6, and B5 of Landsat-8 were composited into false-color images that were taken as the training or testing datasets for DL and ML models.

III. Methodology

A. SVM-Based Burned Area Classification

As a supervised learning algorithm, SVM is a typical classifier that determines the hyperplane (decision boundary) in separating the boundary of each class, for data in d dimensions, as shown in Fig. 2(a). An optimal hyperplane is measured by the distance between the nearest point of each class, and the decision boundary is maximized. The SVM algorithm for linear classification can be expressed as follows:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{d} \max(0, 1 - y_i(w \cdot x_i + b)) \tag{1}$$

subject to the constraint

$$y_i(w \cdot x_i + b) \geq 1 \quad \forall i \tag{2}$$

where w denotes the weights vector, b represents the bias term, C is a regularization parameter controlling the trade-off between maximizing the margin and minimizing the classification error, $\|\cdot\|$ signifies the Euclidean norm, x_i represents the feature vectors, y_i indicates the corresponding class labels, and d denotes the total number of feature vectors. In this study, a linear SVM is responsible for classifying the burned area pixels to reduce the commission errors inside the perimeters that may occur in the HRNet-based segmentation results.

B. HRNet-Based Burned Area Perimeter Segmentation

HRNet has exhibited impressive segmentation results in burned area mapping, particularly for compact fire-disturbed areas, achieving precise delineations and demonstrating its effectiveness [17]. The semantic segmentation model HRNet connects the high-to-low resolution convolution streams in parallel and fuse multiresolution representations repeatedly [23]. The HRNet model can provide intermediate feature representation outputs in different levels, as shown in Fig. 2(b). The structure of HRNet is composed of multiple stages with an increasing number of streams from 1 to 4, corresponding to multiple resolutions, where each stage comprises the

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:30:39 UTC from IEEE Xplore. Restrictions apply.

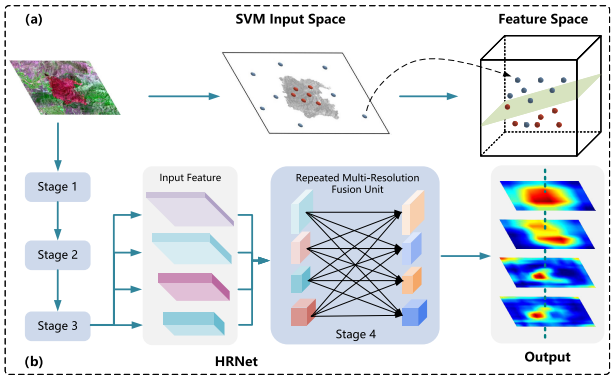

Fig. 2. (a) Schematic of SVM and (b) HRNet processes.

specific streams associated with four different resolutions. This configuration facilitates a finer and more comprehensive representation of features related to the burned areas in different scales. We apply the HRNet model described above to extract the burned area delineations from the segmentation results, finally producing the wildfire fronts, namely, perimeters. However, this region-based segmentation model shows potential commission errors (i.e., unburned areas) inside the perimeters, because inner unburned pixels have a contextual connection with the neighbored burned ones. Conversely, SVM classification can maintain the independent characteristics of burned area pixels. It does not require the use of fixed thresholds that are constrained by geographical conditions and large training datasets, making the ensemble mapping framework applicable. Therefore, as shown in Fig. 2, we managed to adopt an ensemble learning framework combining a pixelwise SVM classification algorithm with a region-based HRNet segmentation model to continuously refine the burned area progression.

C. Ensemble Learning Framework

Initial SVM_0 and HRNet models are separately trained based on training scenarios with the postfire multispectral images and CalFire reference data over 13 major wildfire events. We validate the proposed ensemble learning framework for burned area progression mapping over the Sand fire. The model parameters of the trained HRNet are fixed, but the parameter of SVM will be continually updated. The first multispectral image available in combustion processes ($n = 1$) is represented as E_1. The trained SVM_1 model is similar to SVM_0 at this stage. As shown in Fig. 3, BA_0 and E_0 are set as *NULL*, and SVM_1 is applied to process the input E_1 to obtain $\mathrm{BA}_1^{\mathrm{SVM}}$. $\mathrm{BA}_1^{\mathrm{DL}}$ is the burned area prediction map using HRNet at the first stage. Then, BA_1 is generated from the AND fusion of $\mathrm{BA}_1^{\mathrm{DL}}$ and $\mathrm{BA}_1^{\mathrm{SVM}}$ without the OR operation, since BA_0 is set as *NULL*. For the second available imagery ($n = 2$), SVM_2 will be trained from scratch based on the previous outcome BA_1 as a reference and previous input imagery E_1.

Algorithm 1 summarizes the proposed framework for burned area progression mapping. The input images consist of $E_1, \ldots, E_n, \ldots, E_N$ that are collected during wildfire events, regarding either Sentinel-2 MSI or Landsat-8 OLI imageries across the subsequent acquisition time. Each SVM_n model is

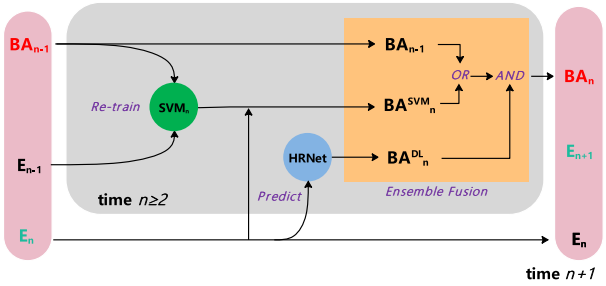

Fig. 3. Near real-time burned area progression mapping framework using ensemble fusion learning.

trained based on the output in the previous event, namely, E_{n-1} and BA_{n-1} as training image and reference data, respectively. E_n is regarded as the present-day input imagery. A trained HRNet model is applied to map the burned area perimeter $\mathrm{BA}_n^{\mathrm{DL}}$. The SVM_n is used to classify the burned area pixels regarding the input E_n, eventually producing the $\mathrm{BA}_n^{\mathrm{SVM}}$. The ensemble fusion step to produce the BA_n consists of an integration of BA_{n-1} and $\mathrm{BA}_n^{\mathrm{SVM}}$ with the logical OR operation first and follows the AND operation for intersection with $\mathrm{BA}_n^{\mathrm{DL}}$ in Fig. 3. The end of the progression depends on the intersection of the burned areas of the two temporal intervals (i.e., BA_n and BA_{n-1}) until the wildfire front grows slowly with a minor expansion.

Algorithm 1 Proposed Ensemble Learning Framework for Burned Area Progression Mapping

Input: Trained model: SVM_0 and HRNet; Raw scenes over fire event from Landsat-8 OLI and Sentinel-2 MSI: $\mathrm{E}_1, \ldots, \mathrm{E}_n, \ldots, \mathrm{E}_N$ for time series from 1 to N.

Output: Burned areas BA_n for imagery input E_n.

Initialize BA_0 and E_0 as *NULL* and $\mathrm{SVM}_1 = \mathrm{SVM}_0$:
for $n \leftarrow 1$ *to* N **do**
  Train SVM_n using BA_{n-1} and E_{n-1} ($n \geq 2$);
  $\mathrm{BA}_n^{\mathrm{DL}} \leftarrow \mathrm{HRNet} + \mathrm{E}_n$;
  $\mathrm{BA}_n^{\mathrm{SVM}} \leftarrow \mathrm{SVM}_n + E_n$;
  $\mathrm{BA}_n \leftarrow \mathrm{BA}_n^{\mathrm{DL}}$ **AND** ($\mathrm{BA}_n^{\mathrm{SVM}}$ **OR** BA_{n-1}) ($n \geq 2$);
  Evaluate the $\mathrm{IoU}_{\mathrm{BA}}$ of BA_n and BA_{n-1} ($n \geq 2$):
  if *no progression* **then**
  end
end

D. Performance Validation

The burned area segmentation results of HRNet are validated through the reference data from CalFire, which provides the fire perimeter of the Sand fire on July 28, 2016. Burned area classification of SVM and the final burned area results of the proposed method are validated in comparison with the reference map using a dNBR-based thresholding approach within the official wildfire perimeters and visual inspection of postfire very-high-resolution satellite image. If there exists fire disturbance between prefire and postfire images, the dNBR

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:30:39 UTC from IEEE Xplore. Restrictions apply.

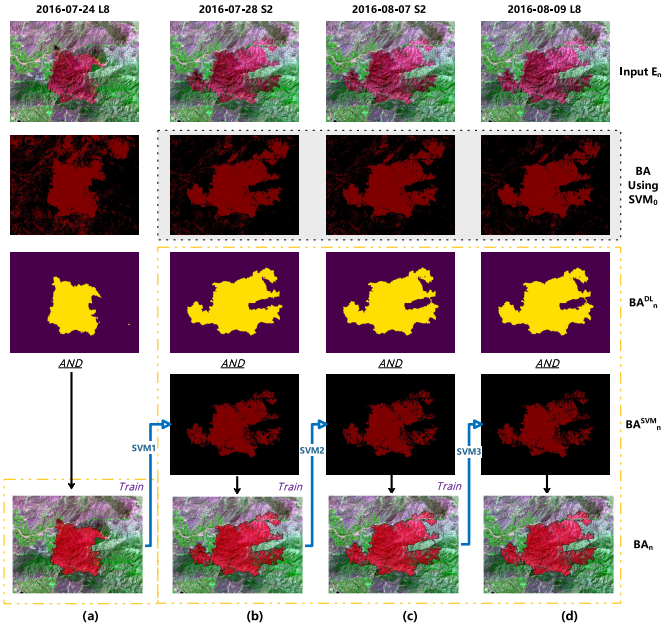

Fig. 4. Burned area progression mapping results over the Sand fire with multispectral data in near real time. S2: Sentinel-2. L8: Landsat-8. (a) $n = 1$. (b) $n = 2$. (c) $n = 3$. (d) $n = 4$.

index generally has a bimodal distribution. Therefore, automatic thresholding methods (e.g., OTSU method [26]) was applied to highlight the burned areas.

IV. Results

A. Qualitative Performance

The results of burned area progression mapping are shown in Fig. 4. During the initiation phase in Fig. 4(a), HRNet generated the initial burned area delineation, denoted as BA_1^{DL}, using input Landsat-8 image E_1 captured in July 24, 2016. Concurrently, burned area detection results using SVM_0 showed high commission errors and limited generalization capabilities, as shown in the second row of Fig. 4(a). The refined burned area mapping result for the first input multispectral imagery E_1, denoted as BA_1, was produced by intersecting the regions identified by BA^{SVM_0} and BA_1^{DL} using the AND operation, as shown in Fig. 4(a). With the arrival of imagery from the second acquisition date ($n = 2$), the previous reference BA_1 and input imagery E_1 were utilized to retrain a new SVM_2 model, enabling the prediction of BA_2^{SVM}. Meanwhile, HRNet could generate the second burned area delineation, denoted as BA_2^{DL}. At the moments ($n = 2$–4), the BA_2, BA_3, and BA_4 are produced by the logical OR operation between previous BA_{n-1} and new BA_n^{SVM}, whose results follow the AND operation for intersection with present-day BA_n^{DL}.

The updated SVM models based on the previous BA_{n-1} results can produce the refined SVM results, namely, BA_2^{SVM}, BA_3^{SVM}, and BA_4^{SVM} in the fourth row in Fig. 4. They show much better performance than the burned area mapping results using SVM_0 alone, as shown in the second row in Fig. 4. This demonstrates that a pixelwise model, such as SVM, performs better on a small-scale dataset with specific land-cover change information rather than a large-scale training dataset over several biomes, as shown in Fig. 1.

TABLE I

Performance (%) of the Proposed Method With Consecutive Imageries E_1, E_2, E_3, and E_4 Acquired During Sand Fire Event

E_n		BA^{SVM_0}	BA_n^{SVM}	BA^{DL}	BA_n(ours)
$n = 1$	Kappa	41.18	41.18	–	**87.34**
	IoU_{BA}	35.13	35.13	–	**80.21**
	IoU_{uBA}	73.89	73.89	–	**96.68**
$n = 2$	Kappa	67.51	81.03	96.77⋆	**84.03**
	IoU_{BA}	60.59	73.72	95.33⋆	**77.51**
	IoU_{uBA}	84.26	92.61	98.34⋆	**93.60**
$n = 3$	Kappa	68.81	80.05	–	**84.42**
	IoU_{BA}	61.88	73.23	–	**78.05**
	IoU_{uBA}	84.95	91.16	–	**93.69**
$n = 4$	Kappa	72.70	79.78	–	**84.99**
	IoU_{BA}	65.69	72.15	–	**78.83**
	IoU_{uBA}	87.13	92.22	–	**93.84**
Mean	Kappa	62.55	70.75	–	**85.19**
	IoU_{BA}	55.82	63.55	–	**78.65**
	IoU_{uBA}	82.55	87.47	–	**94.45**

Note: **BA**-burned area class; **uBA**-unburned area class. ⋆ denotes the perimeter validation between the official Sand fire perimeter and HRNet segmentation result in E_2. – refers to the reference unavailable.

To elaborate further, the front of the wildfire progresses slowly, even to be extinguished visually due to the absence of significant expansion in burned areas from BA_3^{DL} to BA_4^{DL}, as shown in Fig. 4. On the other hand, the relatively low expansion between detected burned areas in consecutive acquisition times indicates minimal overlap, possibly reflecting the subtle changes in fire behaviors during this period (e.g., BA_3 to BA_4 in Fig. 4). These changes mainly come from the slow front progression, emphasizing the significance of accurately capturing the boundary delineation of wildfire spread in near real time for effective fire management and mitigation strategies.

B. Quantitative Performance

Four multispectral imageries over the Sand fire are carried out to evaluate the performance of the proposed ensemble learning framework for burned area progression mapping. The quantitative evaluations demonstrate the superior performance of the proposed method in terms of the Kappa index and IoU, as shown in Table I. The segmentation result BA^{DL} of HRNet highlights the burned area perimeter with a very high Kappa coefficient over 96%, compared with the CalFire reference data for the E_2 imagery on July 28, 2016. Due to the abundant training data, HRNet in this study performs better than that in [17] with a Kappa of 89% regarding the compact burned area mapping for a Mediterranean forest fire. The perimeters generated by HRNet can keep a high level of accuracy, serving effectively as the boundary filter on the SVM-based classification results. Regarding the BA_n, the mean IoU of the burned area class and unburned area class is 86.55%, higher than some typical burned area mapping methods with single postfire Sentinel-2 data, such as threshold-based method LBAP (39.27%), U-Net (79.67%), and Burnt-Net (86.11%) in [20] over 11 major wildfire events across six countries, and U-Net (58.3%) and SegFormer (56.3%) in [27] for 340 wildfire events across California.

The BA^{SVM_0} shows the prediction results with low accuracy based on the initial SVM_0 model. Regarding the BA_n^{SVM}

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:30:39 UTC from IEEE Xplore. Restrictions apply.

results of the updated SVM model, the mean Kappa coefficient improves from 62.55% to 70.75%. Furthermore, the boundary constraint of HRNet-based fire perimeter (i.e., BA^{DL}) significantly improves the burned area detection with a mean Kappa up to 85.19%. On the other hand, the perimeter polygons obtained from HRNet facilitate the detection of final BA_n with an increase by more than 3% over BA_n^{SVM} ($n \geq 2$) in the IoU_{BA}, since burned areas from HRNet can reduce the commission errors that do not fall in the perimeters inside. The ensemble fusion operation can improve the final BA_n step by step, because the BA_{n-1} can compensate the BA_n to reduce the omission errors inside the perimeter.

V. Conclusion

Monitoring large-scale burned area progression is essential for effective wildfire policy in wildfire suppression, emergency response, and hazard planning. Therefore, this letter investigates the potential of multispectral data on near real-time progression mapping using pixelwise SVM classification algorithm and region-based HRNet segmentation model. We proposed a hybrid ensemble learning framework to map the burned areas continuously using Landsat-8 OLI and Sentinel-2 MSI data available. HRNet provides the boundary-constraint delineations for iterative training of SVM models to refine the burned area spatially as new imagery arrives. Specifically, the ensemble fusion step fuses the present-day SVM-based burned areas and previously determined burned areas, followed by the intersection with the present-day HRNet-based delineation. The experiment results over the Sand fire validated the good performance of the perimeter prediction using HRNet and the progression monitoring based on the ensemble framework.

Acknowledgment

The authors would like to express their sincere gratitude to the editors and anonymous reviewers for their insightful comments during the review process of this letter.

References

[1] W. M. Jolly et al., "Climate-induced variations in global wildfire danger from 1979 to 2013," *Nature Commun.*, vol. 6, no. 1, p. 7537, Jul. 2015.

[2] H. Clarke, B. Cirulis, N. Borchers-Arriagada, R. Bradstock, O. Price, and T. Penman, "Health costs of wildfire smoke to rise under climate change," *NPJ Climate Atmos. Sci.*, vol. 6, no. 1, p. 102, Jul. 2023.

[3] K. Luo, X. Wang, M. de Jong, and M. Flannigan, "Drought triggers and sustains overnight fires in north America," *Nature*, vol. 627, no. 8003, pp. 321–327, Mar. 2024.

[4] M. J. Wooster et al., "Satellite remote sensing of active fires: History and current status, applications and future requirements," *Remote Sens. Environ.*, vol. 267, Dec. 2021, Art. no. 112694.

[5] L. Giglio, J. Descloitres, C. O. Justice, and Y. J. Kaufman, "An enhanced contextual fire detection algorithm for MODIS," *Remote Sens. Environ.*, vol. 87, nos. 2–3, pp. 273–282, Oct. 2003.

[6] Y. Chen, C. Lu, X. Huang, S. Xie, and Y. Sun, "Burned area estimation using a new accuracy verification method based on Sentinel-2 images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[7] R. Lasaponara, C. Fattore, and G. Modica, "Imaging burned areas and fire severity in Mediterranean fragmented ecosystems using Sentinel-1 and Sentinel-2: The case study of Tortoli–Ogliastra fire (Sardinia)," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[8] M. A. Crowley, J. A. Cardille, J. C. White, and M. A. Wulder, "Generating intra-year metrics of wildfire progression using multiple open-access satellite data streams," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111295.

[9] R. Lasaponara and B. Tucci, "Identification of burned areas and severity using SAR Sentinel-1," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 917–921, Jun. 2019.

[10] A. Alonso-Benito, P. A. Hernandez-Leal, A. Gonzalez-Calvo, M. Arbelo, and A. Barreto, "Analysis of different methods for burnt area estimation using remote sensing and ground truth data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Aug. 2008, pp. 828–831.

[11] X. Cao, J. Chen, B. Matsushita, H. Imura, and L. Wang, "An automatic method for burn scar mapping using support vector machines," *Int. J. Remote Sens.*, vol. 30, no. 3, pp. 577–594, Feb. 2009.

[12] R. Lasaponara, A. M. Proto, A. Aromando, G. Cardettini, V. Varela, and M. Danese, "On the mapping of burned areas and burn severity using self organizing map and Sentinel-2 data," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 5, pp. 854–858, May 2020.

[13] A. Shama et al., "A burned area extracting method using polarization and texture feature of Sentinel-1A images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[14] J. Zhao, Z. Zhang, S. Han, C. Qu, Z. Yuan, and D. Zhang, "SVM based forest fire detection using static and dynamic features," *Comput. Sci. Inf. Syst.*, vol. 8, no. 3, pp. 821–841, 2011.

[15] F. Zhao, C. Huang, and Z. Zhu, "Use of vegetation change tracker and support vector machine to map disturbance types in greater Yellowstone ecosystems in a 1984–2010 Landsat time series," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1650–1654, Aug. 2015.

[16] O. Zammit, X. Descombes, and J. Zerubia, "Burnt area mapping using support vector machines," *Forest Ecol. Manage.*, vol. 234, p. S240, Nov. 2006.

[17] X. Hu, Y. Ban, and A. Nascetti, "Uni-temporal multispectral imagery for burned area mapping with deep learning," *Remote Sens.*, vol. 13, no. 8, p. 1509, Apr. 2021.

[18] P. Zhang, X. Hu, Y. Ban, A. Nascetti, and M. Gong, "Assessing Sentinel-2, Sentinel-1, and ALOS-2 PALSAR-2 data for large-scale wildfire-burned area mapping: Insights from the 2017–2019 Canada wildfires," *Remote Sens.*, vol. 16, no. 3, p. 556, Jan. 2024.

[19] W. Fang, Y. Fu, and V. S. Sheng, "Dual backbone interaction network for burned area segmentation in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[20] S. T. Seydi, M. Hasanlou, and J. Chanussot, "Burnt-Net: Wildfire burned area mapping with single post-fire Sentinel-2 data and deep learning morphological neural network," *Ecolog. Indicators*, vol. 140, Jul. 2022, Art. no. 108999.

[21] P. Zhang, Y. Ban, and A. Nascetti, "Learning U-Net without forgetting for near real-time wildfire monitoring by the fusion of SAR and optical time series," *Remote Sens. Environ.*, vol. 261, Aug. 2021, Art. no. 112467.

[22] Y. Ban, P. Zhang, A. Nascetti, A. R. Bevington, and M. A. Wulder, "Near real-time wildfire progression monitoring with Sentinel-1 SAR time series and deep learning," *Sci. Rep.*, vol. 10, no. 1, p. 1322, Jan. 2020.

[23] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[24] E. Dinerstein et al., "An ecoregion-based approach to protecting half the terrestrial realm," *BioScience*, vol. 67, no. 6, pp. 534–545, Jun. 2017.

[25] R. Chastain, I. Housman, J. Goldstein, M. Finco, and K. Tenneson, "Empirical cross sensor comparison of Sentinel-2A and 2B MSI, Landsat-8 OLI, and Landsat-7 ETM+ top of atmosphere spectral characteristics over the conterminous United States," *Remote Sens. Environ.*, vol. 221, pp. 274–285, Feb. 2019.

[26] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.

[27] D. R. Cambrin, L. Colomba, and P. Garza, "CaBuAr: California burned areas dataset for delineation [software and data sets]," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 3, pp. 106–113, Sep. 2023.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:30:39 UTC from IEEE Xplore. Restrictions apply.